Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
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ChessMate: AI Based Smart Chess Board

Project Proposal
Course Title: Artificial Intelligence Lab
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Lab Project Status	
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Chapter 1

Introduction

1.1 Overview

The AI Chess Board is an intelligent system developed using Python programming language that simulates a real chess game between a human player and a computer. This project is inspired by modern chess engines and online platforms where users can play chess against artificial intelligence. The main purpose of this system is to implement game logic along with decision-making capability so that the computer can play chess intelligently.

In this system, the user can play chess by making moves, while the AI responds with calculated moves based on a decision-making algorithm. The program evaluates different possible moves and selects the best one using techniques like Minimax Algorithm and Alpha-Beta Pruning. These algorithms help the AI to predict future game states and make optimal decisions.

The project is developed in Python because of its simplicity and powerful libraries. Python allows easy implementation of complex algorithms and data structures required for game development. The system can be run on any modern computer and provides a simple interface for interaction, making it suitable for both learning and demonstration purposes.

1.2 Motivation

Chess is one of the most popular strategy games in the world and is widely used to study artificial intelligence and problem-solving techniques. Many advanced chess engines are built using complex algorithms, but students often find it difficult to understand how these systems actually work internally.

This project is motivated by the desire to understand how AI makes decisions in a structured environment like a chess game. While high-level applications hide most of the logic behind libraries and frameworks, this project focuses on building the core logic from scratch using Python.

I chose this project because it combines game development with artificial intelligence concepts. It helps in understanding how algorithms like Minimax work in real scenarios. This project also improves logical thinking, algorithm design, and problem-solving skills. By building this system, I aim to gain a deeper understanding of how intelligent systems operate and how decision-making processes are implemented in software.

1.3 Problem Definition

1.3.1 Problem Statement

In a traditional chess game, players rely on their thinking ability to predict moves and make strategic decisions. However, designing a system where a computer can play chess intelligently is a challenging task.

The problem this project addresses is how to implement an AI-based chess system using Python that can simulate intelligent gameplay. The system must handle board representation, legal move generation, and decision-making using algorithms.

The main challenge is to evaluate multiple possible game states efficiently and choose the best move within a reasonable time. This requires proper use of algorithms, recursion, and optimization techniques such as Alpha-Beta Pruning.

1.3.2 Complex Engineering Problem

Based on the problem and motivation discussed above, the following table summarizes the key engineering attributes and how they are addressed in this project.

1.4 Design Goals/Objectives

The main objectives of this project are:

- To develop an AI-based chess game using Python.
- To implement decision-making algorithms such as Minimax and Alpha-Beta Pruning.
- To simulate intelligent gameplay between a human and a computer.
- To understand and apply artificial intelligence concepts in a practical scenario.
- To improve problem-solving and algorithm design skills.

1.5 Application

This project has several practical applications:

Table 1.1: Summary of the attributes touched by the mentioned projects

Name of the P Attributes	Explain how to address
P1: Depth of knowledge required	The project requires understanding of data structures, recursion, game trees, and AI algorithms such as Minimax and Alpha-Beta Pruning. Knowledge of Python programming is also essential.
P2: Range of conflicting requirements	The system must balance between accuracy and performance. A deeper search gives better decisions but increases computation time, so optimization is necessary.
P3: Depth of analysis required	The system needs to analyze multiple future game states and evaluate them using a scoring function. Efficient decision-making requires proper analysis of possible moves.
P4: Familiarity of issues	Common issues include incorrect move generation, inefficient recursion, and performance limitations. The project helps identify and solve these problems.
P5: Extent of applicable codes	The entire system is implemented using Python without relying on external chess engines. All logic such as move validation and decision-making is manually implemented.
P6: Extent of stakeholder involvement and conflicting requirements	The project is mainly academic. The stakeholders (student and instructor) expect correctness, clarity, and proper implementation of AI concepts.
P7: Interdependence	Different components such as board representation, move generation, and AI decision-making are highly interdependent. Failure in one component affects the entire system.

1.5.1 Educational Use

This system can be used as a learning tool for students studying artificial intelligence, algorithms, and game development. It helps visualize how decision-making algorithms work.

1.5.2 Game Development

The project provides a foundation for developing more advanced games with AI features. It can be extended with graphical interfaces and improved algorithms.

1.5.3 AI Research and Practice

Chess is widely used in AI research. This project helps beginners understand fundamental AI concepts and prepares them for more advanced topics.

1.5.4 Skill Development

Working on this project improves programming skills, logical thinking, and understanding of complex systems. It is useful for students preparing for careers in software development and artificial intelligence.